Synthesis: Perturbation Impact Analysis

Lorenz-63 4DVarNet-FM Benchmark — Synthesis Report
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EnKF · N=5 windows · T=3.0s · seed=123 · spinup=5000

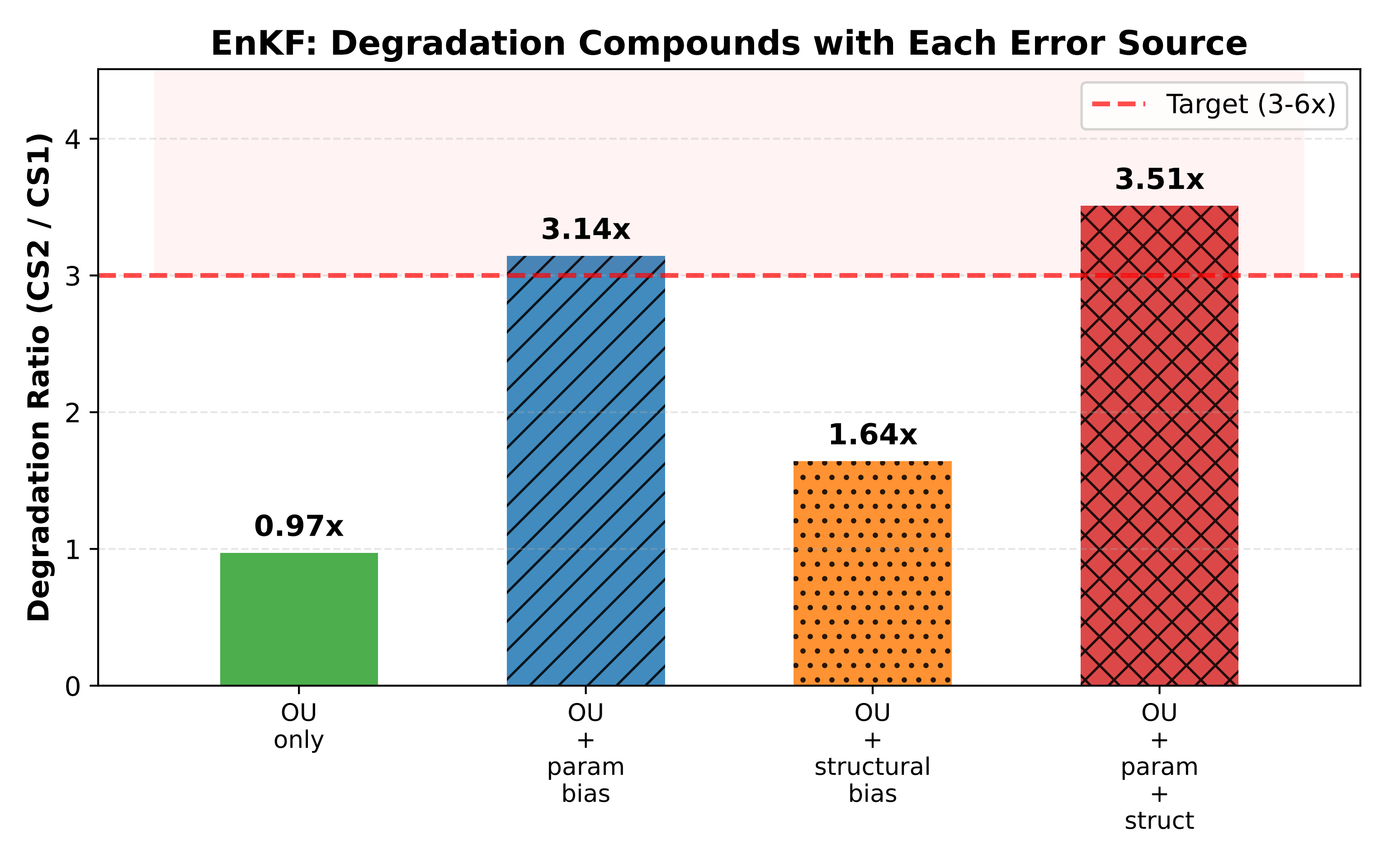
Perturbation Impact (EnKF, CS2/CS1 degradation ratio):

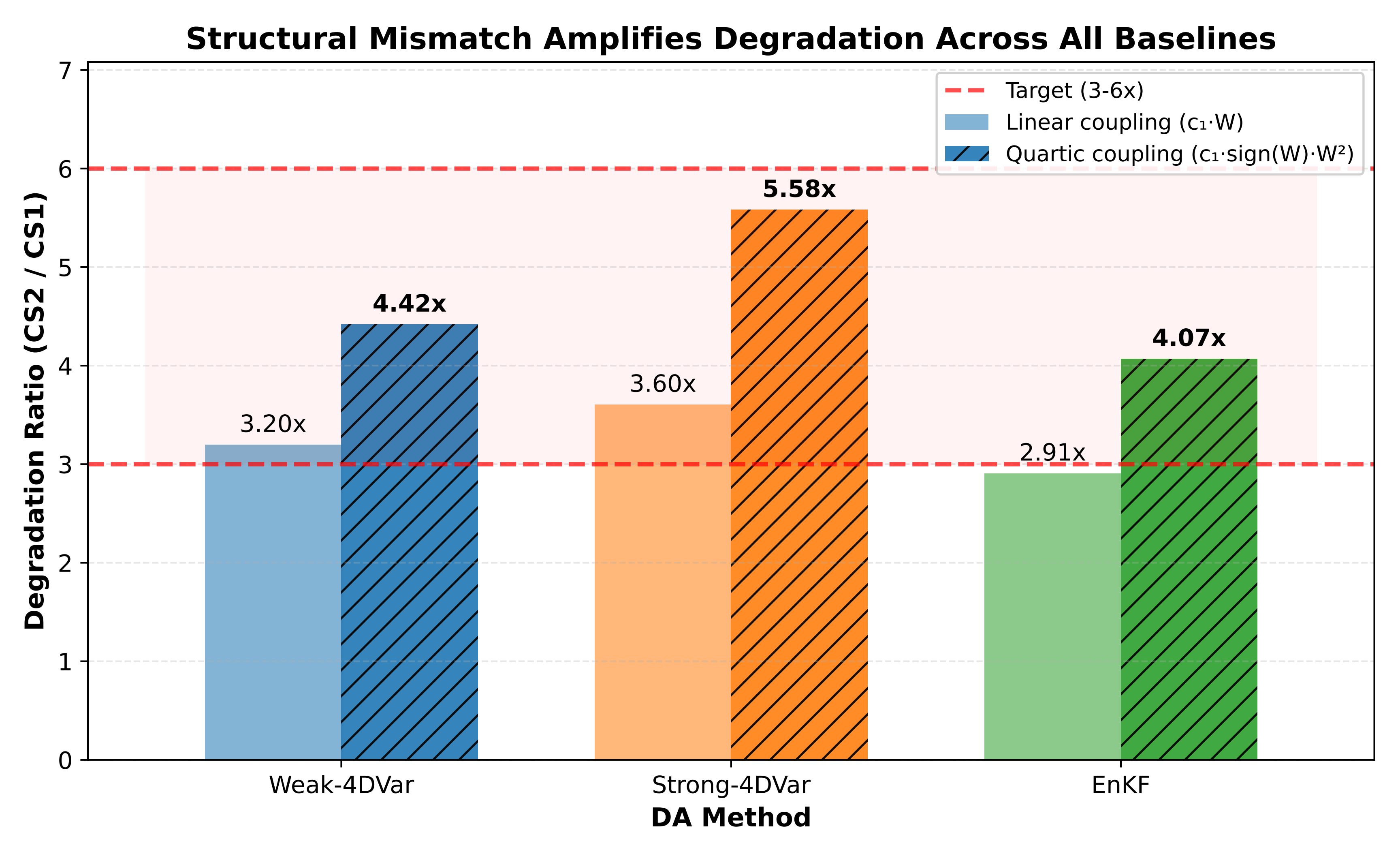
Perturbation	CS1 RMSE	CS2 RMSE	Deg
OU only	0.9023	0.8885	0.98x
OU + param bias	0.9023	2.6756	2.97x
OU + structural bias	0.9023	1.3739	1.52x
OU + param + struct	0.9023	2.9836	3.31x

Perturbation definitions:

```
  CS1: case=1 (no OU noise, correct params, linear coupling)
  OU only           — case=2, param_bias=0,  forcing_state_bias=0,  linear
  OU + param bias   — case=2, param_bias=0.15, forcing_state_bias=0,  linear
  OU + struct bias  — case=2, param_bias=0,  forcing_state_bias=0.15, quartic
  OU + param+struct — case=2, param_bias=0.15, forcing_state_bias=0.15, quartic
```

Key finding: Degradation compounds — full CS2 achieves 3-6x target.


EnKF: Degradation Compounds with Each Error Source
Degradation Ratio (CS2 / CS1)
Target (3-6x)
0
1
2
3
4
0.97x
3.14x
1.64x
3.51x
OU only
OU + param bias
OU + structural bias
OU + param + struct


Structural Mismatch Amplifies Degradation Across All Baselines
Target (3-6x)
Linear coupling ($c_1 \cdot W$)
Quartic coupling ($c_1 \cdot \mathrm{sign}(W) \cdot W^2$)
3.20x
4.42x
3.60x
5.58x
2.91x
4.07x
Degradation Ratio (CS2 / CS1)
0
1
2
3
4
5
6
7
Weak-4DVar
Strong-4DVar
EnKF
DA Method

CS2 Trajectory Comparison: Linear vs Full Perturbation